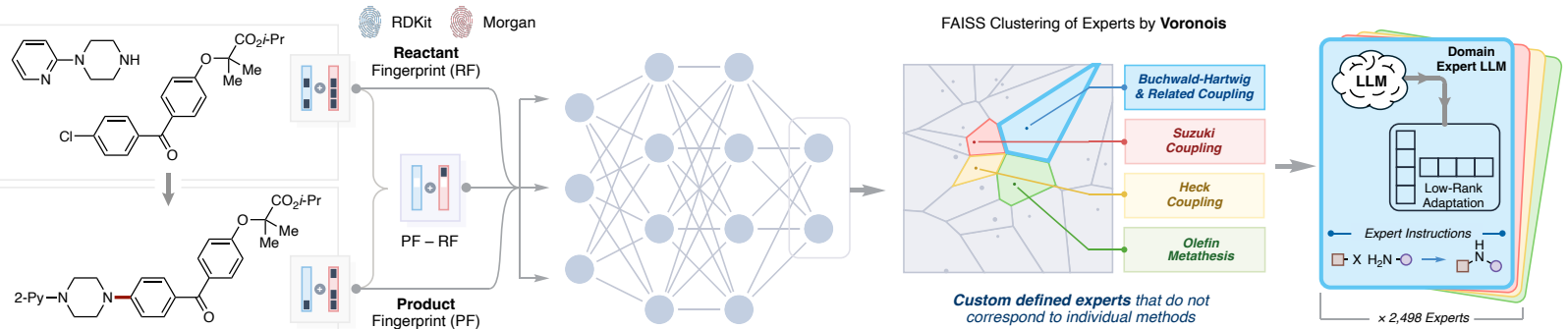




Expert Knowledge Composition (an example)

Expert ID: 883 Centroid distances: 86 (Confidence)

Chloro Buchwald-Hartwig amination: 70.69%

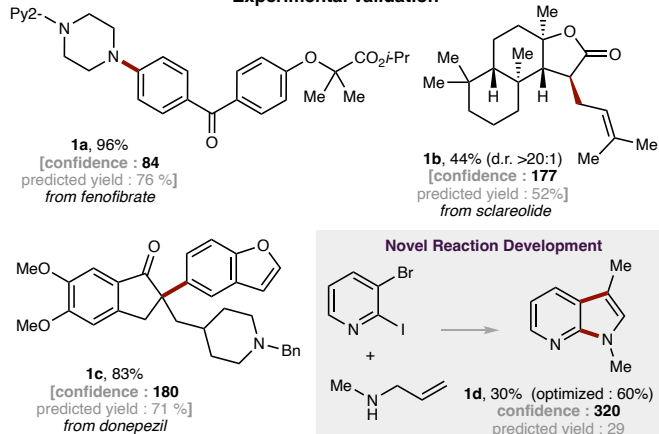
Chloro S_NAr reaction: 17.22%

Chloro Goldberg reaction and related reactions: 0.26%

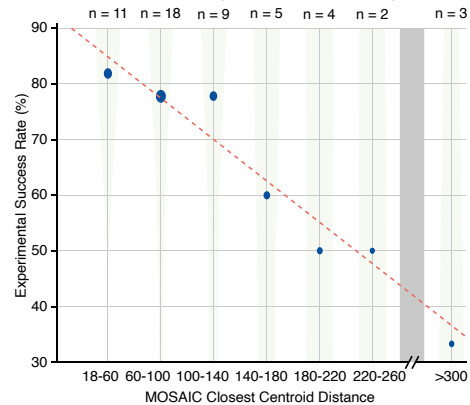
Prediction

To a stirred solution of isopropyl 2-(4-(4-chlorobenzoyl)phenoxy)-2-methylpropanoate (**0.50 g, 1.41 mmol**) in 1,4-dioxane (**20 mL**) was added 2-(piperazin-1-yl)pyridine (**0.26 g, 1.69 mmol**) and cesium carbonate (**1.38 g, 4.2 mmol**) at room temperature. The reaction mixture was purged with argon for 20 min. Then $Pd_2(dba)_3$ (**0.13 g, 0.14 mmol**) and X-phos (**0.14 g, 0.14 mmol**) were added to the reaction mixture and refluxed to 110° C. for 16 h. After completion of the reaction (monitored by TLC), the reaction mixture was filtered through a pad of celite and evaporated to get the crude product. The crude product was purified by silica gel column chromatography using 20% EtOAc-hexane to afford isopropyl 2-methyl-2-(4-(4-(pyridin-2-yl)piperazin-1-yl)benzoyl)phenoxy)propanoate (**0.60 g, 1.16 mmol, 82% yield**) as a **yellow solid**.

Experimental validation



MOSAIC confidence & experimental success rate plot



Establishes correlation between MOSAIC confidence & feasibility